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ARTICLE

Estimation of the number of people with Down syndrome in Latin America and the Caribbean

Gert de Graaf[1], Frank Buckley[2], Brian G. Skotko[3,4,*]

[1]Dutch Down Syndrome Foundation, Meppel, The Netherlands; [2]Down Syndrome Education International, Kirkby Lonsdale, Cumbria, United Kingdom; [3]Down Syndrome Program, Division of Medical Genetics and Metabolism, Department of Pediatrics, Massachusetts General Hospital, Boston, MA; [4]Department of Pediatrics, Harvard Medical School, Boston, MA

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ABSTRACT

Purpose: We estimate the live births and overall population with Down syndrome (DS) in Latin America and the Caribbean from 1950 to 2020. This study adds to previous work from the United States, Europe, Australia, New Zealand, and Canada.
Methods: We used a validated model, adjusted for the region, to estimate live birth prevalence, population prevalence, and life expectancy of people with DS, utilizing data from the UN World Population Prospects and other sources.
Results: We estimate 18,655 (95% CI: 18,387-18,923) annual live births of children with DS (2016-2020) in Latin America and the Caribbean (18.4 per 10,000 live births, 95% CI: 18.2-18.7). The estimated total population of people with DS was 574,092 (95% CI: 572,607-575,577) in 2020 (8.8 per 10,000 inhabitants, 95% CI: 8.78-8.83). Mean life expectancy increased from 10 years in 1950 to 48 years in 2020.
Conclusion: In Latin America and the Caribbean, the overall population of people with DS is still growing significantly. This is due to sharply increasing survival rates for children with DS in recent decades, combined with relatively few selective abortions in this region.

Introduction

The epidemiology of Down syndrome (DS) has been changing around the world in recent years.[1-6] Advances in medicine and surgical procedures have extended the average lifespan for people with DS. For example, in the United States, the life expectancy for individuals with DS significantly increased from a mean of 26 years (median 4 years) in 1950 to a mean of 53 years (median 58 years) by the 2010s.[1] More people are also waiting until they are older to become pregnant, increasing the number of fetal conceptions with DS. At the same time, in many countries, more expectant persons have access to prenatal cell-free DNA screening, which widens their pregnancy options.

In some countries, the number of selective terminations for DS now outpaces the mathematically boosting effects of advanced maternal age and increased life spans for people

*Correspondence and requests for materials should be addressed to Brian Skotko, Massachusetts General Hospital, 125 Nashua Street, Suite 840, Boston, MA 02118. *Email address:* bskotko@mgh.harvard.edu

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with DS. In short, the population of people with DS has begun to contract in some countries: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Greece, Iceland, Italy, The Netherlands, New Zealand, Norway, Portugal, Spain, Switzerland, and the United Kingdom.[3,4] In other places, such as many Eastern Europe countries, the population of people with DS is clearly expanding.[3] In most of these countries, prenatal screening and selective terminations have only recently become more available, and the survival of newborns with DS has climbed rapidly from 1990 onward, with later deinstitutionalization efforts and improved access to medical care. The trend in the United States and Canada falls between the 2 extremes: the overall population for DS is still increasing—but the pace is slowing down, in Canada to almost 0.[1,5]

In this manuscript, we expand our accounting to 50 countries (or territories) in Latin America and the Caribbean. The majority of the countries (or territories) in this region outlaw selective terminations for DS. Only in some of these countries (which are mostly small countries) are prenatal testing and selective terminations available, sometimes for decades, as is the case in Cuba. We expect then that only for some Latin American and Caribbean countries (or territories) will selective terminations have an effect on the overall population for DS. For the region as a whole, and for the larger subregions within this region (the Caribbean, Central America and Mexico), and South America), this effect will be small. In this manuscript, we also construct a more precise survival curve for DS to fit the circumstances of the Latin American and Caribbean countries (or territories), which historically have less resources than countries previously included in modeling. With overall more precise population estimations, governments and their people can begin to have more grounded conversations about their citizens with DS.

Materials and Methods

To estimate the population size of people with DS in countries (or territories) in Latin America and the Caribbean, we used a method that has previously been successfully applied for the United States, Europe, Australia, New Zealand, and Canada.[1-5] We made some adjustments in estimating survival (described in Supplemental Methods 2A).

Estimates of nonselective live births (LBs) of children with DS

In the absence of DS-specific selective terminations, the number of potential liveborn children with DS (nonselective LBs of children with DS) can be estimated by applying a model for the maternal age-specific probability of a live birth with DS to data on the age of mothers in the general population.[7] Details of the sources and procedures can be found in Supplemental Methods 1A.

Estimates of actual LBs of children with DS

We assumed that in countries (or territories) in which termination is not permitted in cases of fetal anomaly, there will be no (or only very few) selective abortions related to DS. We verified whether our estimates of nonselective prevalence for DS align with counts of liveborn children in several of these South American countries.[8-10] This was the case (Supplemental Methods 1B).

There are 18 countries/territories (out of 50) in the region that allow elective terminations for fetal impairment.[11] Only 4 (including 1 territory) have a relatively large population (Cuba, Puerto Rico, Panama, and Mexico). Details of our assumptions on reduction of LBs of children with DS in these 18 countries/territories can be found in Supplemental Methods 1B.

Modeling survival

For each country (or territory) and each birth year, we have created survival curves for individuals with DS. In doing so, we have built upon the earlier work of de Graaf et al[1-5] (Supplemental Methods 2A). However, we have made several adjustments (described in detail in Supplement Methods 2A):

- We used the same set of historical studies on survival in children with DS that de Graaf et al[1-5] used. However, instead of looking at the relation between year of birth and 1-year survival (and later on adapting for differences between countries in child mortality within the general population), we looked directly at the relation between 1-year survival in the general population and 1-year survival in people with DS. We then searched for a prediction model that fit the data well (details are in Supplemental Methods 2A), which we subsequently used to model 1-year survival in people with DS by year and country/territory.
- In earlier studies, we fitted a linear relation between 1-year survival in people with DS and 5-year and 10-year survival, respectively. In a recent study, de Graaf et al[5] concluded that quadratic equations had a slightly higher fit. In this study, we used the quadratic equations.
- In comparison with earlier studies, we have assumed a better survival between 10 to 30 years in countries/territories with a 1-year survival >80% for the corresponding year of birth. The same has been done in the recent study on Canada.[5]

Validating the model

We have checked how well the model's predictions (the original model and the new model with the aforementioned adjustments) align with empirical data. First, we examined counts of people with DS and their distribution across age groups (Supplemental Methods 3A). For Latin America and the Caribbean, these were available from Argentina (as of 2023) and Cuba (as of 2002).[12,13] In addition, we made comparisons with data from 4 countries outside Latin America and the Caribbean. The United Kingdom, France, Australia, and the United States had either recent and likely fairly complete data on people with DS by age group, or recent data based on large samples.[4,14-17]

A second way to validate the model is by comparing age at death data of people with DS from national statistics with the model's predictions.[18] Deceased individuals with DS may not always be registered as having died with DS as the primary cause of death. Assuming that "under-registration" is not dependent on age, we can look at the age distribution of deaths among people with DS (Supplemental Methods 3B). In total, 14 countries (or territories) in Latin America and the Caribbean had sufficient data. Additionally, we examined whether the model works well in this regard for the 4 comparison countries (United Kingdom, France, Australia, and the United States).

Results

LBs and LB prevalence of DS

For Latin America and the Caribbean, between 2016 to 2020, we estimate 18,655 (95% CI: 18,387-18,923) annual LBs of children with DS, and a LB prevalence of 18.4 per 10,000 live births (95% CI: 18.2-18.7). Without elective terminations, we estimate there would have been 18,921 (95% CI: 18,651-19,191) births annually and a LB prevalence around 18.7 per 10,000 live births (95% CI: 18.4-19.0). As such, the estimated reduction of LB prevalence by elective terminations was, on average, 1.4% (95% CI: 0%-3.4%), varying between 0% in countries where termination is not permitted in cases of fetal anomaly to an estimated 69% in Cuba (95% CI: 59%-79%) (Figures 1 and 2, Supplemental Findings 1 and 2).

Among the 3 Latin American and Caribbean regions, nonselective and actual LB prevalence was highest in South America (19.7 per 10,000 LBs (95% CI: 19.4-20.1) and 19.7 per 10,000 LBs (95% CI: 19.3-20.0), respectively), followed by the Caribbean (19.3 per 10,000 LBs (95% CI: 18.3- 20.3) and 17.3 per 10,000 LBs (95% CI: 16.3-18.2), respectively), and Central America and Mexico (16.4 per 10,000 LBs [95% CI: 16.0- 16.9] and 16.1 per 10,000 LBs [95% CI: 15.6-16.5], respectively) (Figures 1 and 2; Supplemental Findings 2). Reduction percentage was highest in the Caribbean with an estimated 10% (95% CI: 3%-17%), followed by Central America and Mexico (2%, 95% CI: 0%-6%), and South America (0.1%, 95% CI: 0%-3%). The relatively high reduction in the Caribbean is weighted by the significant reduction rate in Cuba (69%), a country with a relatively large population within this region.

In Figure 2, we compare Latin America and the Caribbean with various other geographical regions on the development of the LB prevalence of children with DS over time. Latin America and the Caribbean exhibit a different pattern compared with the 5 comparison regions: the United States, Canada, Australia, New Zealand, and Europe (excluding the former East Bloc).[3-6,19] In the 5 comparison regions, we observe a decline in LB prevalence after World War II (WWII) due to a reduction in family size and the trend of having the first child at a younger age. The nadir occurs in the late 1970s or, in Europe, the early 1980s. Subsequently, as a result of delayed childbearing, we see a sharp increase in nonselective LB prevalence across all 5 comparison regions. However, because of selective abortions, the actual LB prevalence decreases in the same period in Australia, New Zealand, and Europe or increases only slightly in the United States and Canada. In recent years, the reduction of LBs of children with DS is as high as 60%-70% in Australia, New Zealand, and Europe, around 55% in Canada and around 37% in the United States[1-6,19]

In Latin America and the Caribbean, the LB prevalence in 1946 to 1950 is significantly higher than in the 5 comparison regions. In contrast to the comparison regions, LB prevalence continues to rise after WWII, only beginning to decline in the early 1970s. The nadir is reached between 2006 and 2010. Only after this period does the nonselective LB prevalence begin to increase. Because of the very limited impact of selective abortions in Latin America and the Caribbean, the actual LB prevalence also increases in recent years. This general pattern over time is observed across the 3 distinctive regions within Latin America and the Caribbean, with minor differences in timing (Figure 2).

Population numbers and population prevalence of DS

The estimated number of people with DS in Latin America and the Caribbean has increased from 52,833 (95% CI: 52,382-53,284) in 1950 (corresponding to a population prevalence of 3.1 per 10,000 inhabitants, 95% CI: 3.11-3.17) to 574,092 (95% CI: 572,607-575,577) in 2020 (8.8 per 10,000 inhabitants, 95% CI: 8.78-8.83) (Figure 3). For estimates by country/territory (as of 2020), see Table 1 and Supplemental Findings 1. The 3 regions within Latin America and the Caribbean show a highly similar trend in the population prevalence of DS (Figure 3, Supplemental Findings 2). In the Caribbean, the estimated number of people with DS was 5706 (95% CI: 5558-5854) in 1950 (3.3 per 10,000 inhabitants, 95% CI: 3.2-3.4) and 33,298 (95% CI: 32,940-33,656) in 2020 (7.6 per 10,000 inhabitants, 95% CI: 7.5-7.7). In Central America and Mexico, these values were 9943 (95% CI: 9748-10,138) (2.6 per 10,000 inhabitants, 95% CI: 2.6-2.7) in 1950 and 159,922 (95% CI: 159,138-160,706) (9.1 per 10,000 inhabitants, 95% CI: 9.02-9.11) in 2020; in South America, 37,185 (95% CI: 36,807-37,563) (3.3 per 10,000

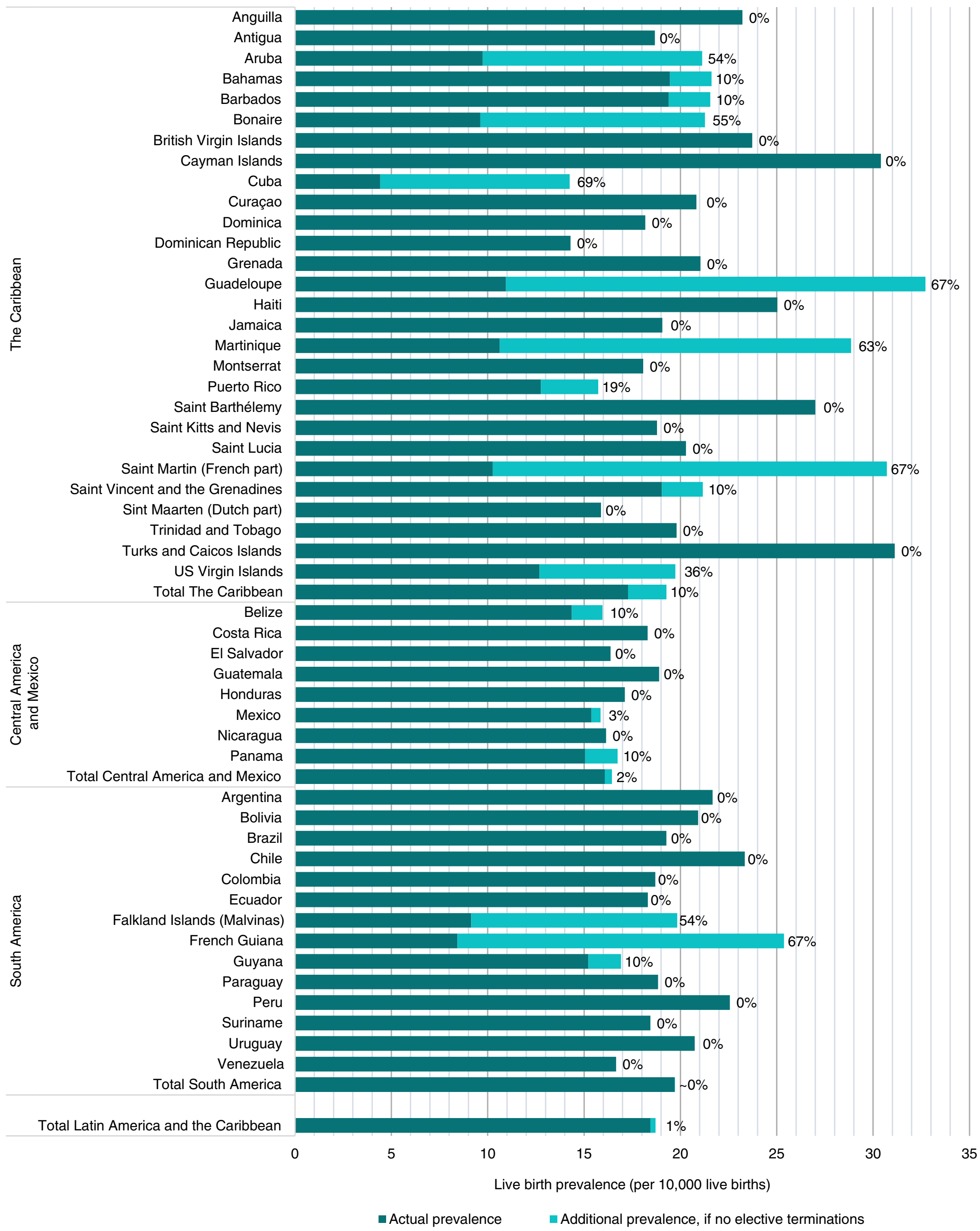

Figure 1 Live birth prevalence estimates of people with Down syndrome (DS) per 10,000 live births (2016-2020) and the effect of elective terminations. Percentages denote the reduction due to DS-specific elective terminations.

inhabitants, 95% CI: 3.24-3.31) in 1950 and 380,872 (95% CI: 379,662-382,082) (8.8 per 10,000 inhabitants, 95% CI: 8.80-8.85) in 2020. For comparison, the DS population prevalence was estimated at 6.7 per 10,000 inhabitants in the United States (as of 2018), 6.3 per 10,000 inhabitants in Europe (excluding the former East Bloc, as of 2015), 6.0 per 10,000 in New Zealand (as of 2020), 5.9 per 10,000 inhabitants in Canada (as of 2020), and 5.2 per 10,000 inhabitants in Australia (as of 2020).[1,3-5,19]

In Latin America and the Caribbean, the effect of DS-related selective terminations on the number of people with DS is minimal (Figure 3). In the absence of selective terminations, the overall population of people with DS, as of 2020, would have been 5129 individuals greater, distributed as follows: 3578 in the Caribbean (of which 2725 would be in Cuba), 1331 in Central America and Mexico (of which 1162 would be in Mexico), and 220 in South America (of which 192 would be in French Guiana). This implies that the total population of people with DS in Latin America, as of 2020, was reduced by approximately 1% (95% CI: 0.9%-1.3%) because of selective terminations, with reductions of 10% (95% CI: 8%-11%) in the Caribbean, 1% (95% CI:

Figure 2 **Estimates of LB prevalence of children with DS per 10,000 LBS: Nonselective and actual.** The data for the United States, Canada, Australia, New Zealand, and Europe were derived from earlier studies.[6] DS, Down syndrome.

0.1-1.5%) in Central America and Mexico, and 0.06% (95% CI: 0%-0.5%) in South America. For comparison, this value was estimated at 23% in the United States (as of 2018), 28% in Canada (as of 2020), 32% in Europe (excluding the former East Bloc and as of 2015), 37% in New Zealand (as of 2020), and 42% in Australia (as of 2020).[1,3-5,19]

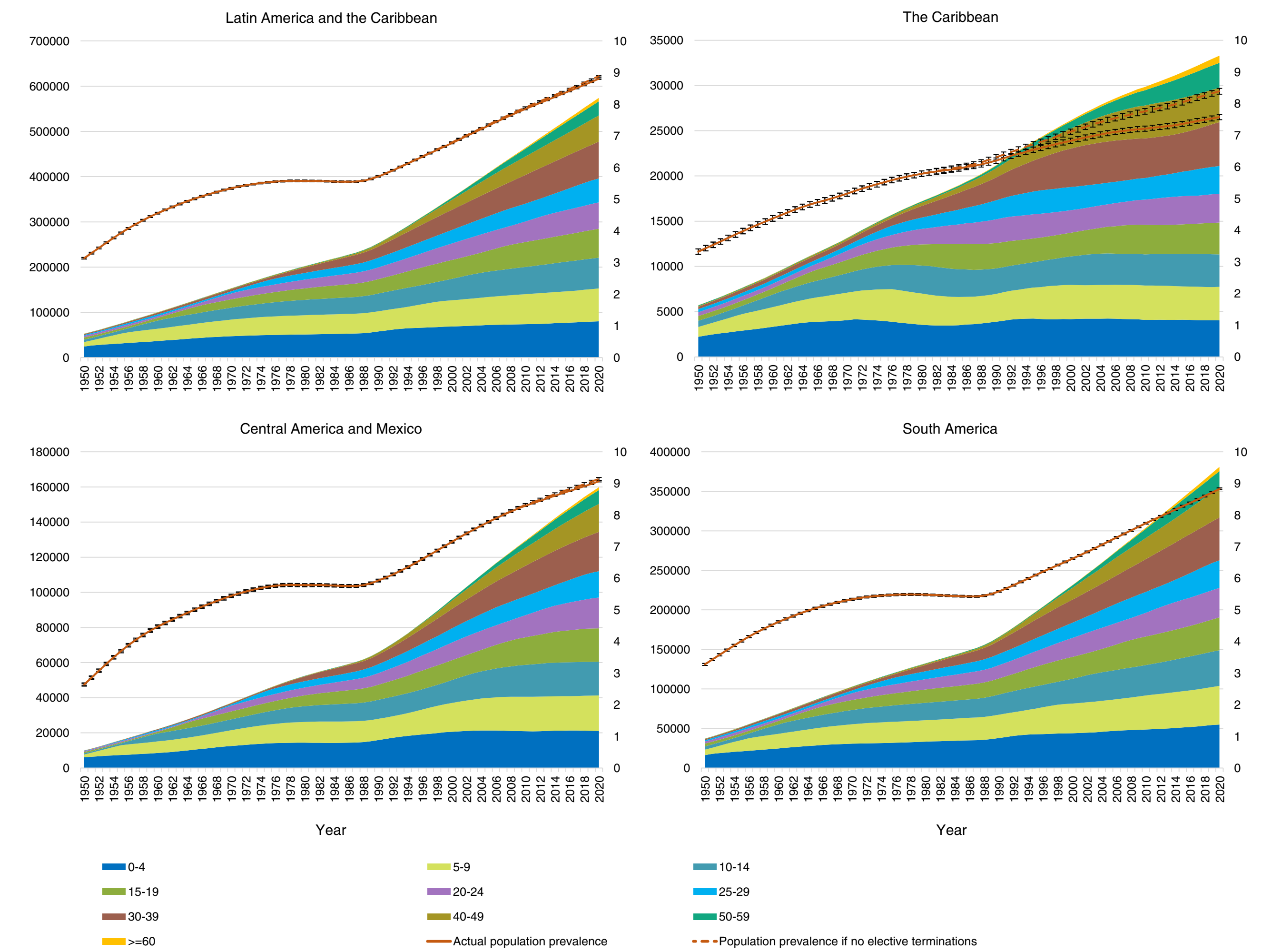

Figure 3 **Estimates of the number of people with DS in Latin America and the Caribbean by age group, 1950-2020.** DS, Down syndrome.

Changes in life expectancy

In Latin America and the Caribbean, the estimated mean life expectancy in DS increased from 10 years in 1950 to 48 years in 2020 (median 0 and 57 years, respectively). The gap between this region and the United States (mean life expectancy of 54, and median of 60 in 2020) is gradually narrowing (Supplemental Methods 2C).

Validation

The results of our validation approach are discussed in detail in Supplemental Methods 3. The adjusted model shows a better fit with the empirical data than the original.

Discussion

LB prevalence of DS

In Latin America and the Caribbean, the LB prevalence of DS has followed a different pattern compared with our comparison areas (Figure 2). As a consequence of higher maternal ages, in Latin America and the Caribbean, LB prevalence of DS started at a higher level after WWII and continued to rise until around 1970. In contrast, in the comparison areas, LB prevalence began at a lower level and remained relatively stable (in the United States and Canada) until 1965 before declining, or it began to decrease as early as the 1950s (in the other 3 areas). This decrease was driven by a reduction in family size and the trend of having the first child at a younger age. From the late 1970s onward, or from the early 1980s in Europe, nonselective LB prevalence of DS began to rise again, because of delayed childbearing. In Latin America and the Caribbean, this increase in nonselective LB prevalence of DS only began around 2010.

In the 5 comparison areas, selective abortions have had a significant impact from the late 1970s onward, with the gap between nonselective and actual LB prevalence of DS widening (Figure 2). As a result, in Australia, New Zealand, and Europe, the actual LB prevalence in 2020 is lower than it was in 1980. In the United States and Canada, the actual LB prevalence of DS has increased since 1980, although at a much slower pace than the rise in nonselective prevalence. In contrast, in Latin America and the Caribbean as a whole,

Table 1 2020 estimates of the number and population prevalence (per 10,000 inhabitants) of people with Down syndrome in Latin American and Caribbean countries (or territories)

Region/Country (or Territory)	Number (95% CI)	Population Prevalence (95% CI)
The Caribbean	33,298 (32,940-33,656)	7.6 (7.5-7.7)
Anguilla	12 (5-19)	7.8 (3.4-12.2)
Antigua and Barbuda	95 (76-114)	10.2 (8.1-12.3)
Aruba	67 (51-83)	6.3 (4.8-7.8)
Bahamas	367 (329-405)	9.0 (8.1-9.9)
Barbados	261 (229-293)	9.3 (8.2-10.4)
Bonaire	15 (7-23)	5.6 (2.8-8.4)
British Virgin Islands	18 (10-26)	5.9 (3.2-8.6)
Cayman Islands	46 (33-59)	6.9 (4.9-8.9)
Cuba	6044 (5892-6196)	5.3 (5.2-5.4)
Curaçao	212 (183-241)	11.2 (9.7-12.7)
Dominica	93 (74-112)	12.9 (10.3-15.5)
Dominican Republic	7516 (7346-7686)	6.8 (6.6-7.0)
Grenada	166 (141-191)	13.4 (11.4-15.4)
Guadeloupe	399 (360-438)	10.1 (9.1-11.1)
Haiti	9942 (9747-10,137)	8.8 (8.6-9.0)
Jamaica	2906 (2800-3012)	10.3 (9.9-10.7)
Martinique	395 (356-434)	10.7 (9.6-11.8)
Montserrat	7 (2-12)	15.7 (4.1-27.3)
Puerto Rico	2922 (2816-3028)	8.9 (8.6-9.2)
Saint Barthélemy	3 (0-6)	3.5 (0-7.5)
Saint Kitts and Nevis	50 (36-64)	10.5 (7.6-13.4)
Saint Lucia	200 (172-228)	11.1 (9.6-12.6)
Saint Martin (French part)	31 (20-42)	9.4 (6.1-12.7)
Saint Vincent and the Grenadines	122 (100-144)	11.7 (9.6-13.8)
Sint Maarten	27 (17-37)	6.3 (3.9-8.7)
Trinidad and Tobago	1229 (1160-1298)	8.1 (7.6-8.6)
Turks and Caicos Islands	42 (29-55)	9.6 (6.7-12.5)
US Virgin Islands	109 (89-129)	10.8 (8.8-12.8)
Central America and Mexico	159,922 (159,138-160,706)	9.1 (9.06.-9.14)
Belize	320 (285-355)	8.1 (7.2-9.0)
Costa Rica	4771 (4636-4906)	9.3 (9.0-9.6)
El Salvador	6779 (6618-6940)	10.8 (10.5-11.1)
Guatemala	18,124 (17,860-18,388)	10.4 (10.2-10.6)
Honduras	10,033 (9837-10,229)	9.9 (9.7-10.1)
Mexico	110,715 (110,063-111,367)	8.8 (8.7-8.9)
Nicaragua	5861 (5711-6011)	8.7 (8.5-8.9)
Panama	3320 (3207-3433)	7.7 (7.4-8.0)
South America	380,872 (379,662-382,082)	8.8 (8.77-8.83)
Argentina	45,134 (44,718-45,550)	10.0 (9.9-10.1)
Bolivia	11,321 (11,112-11,530)	9.5 (9.3-9.7)
Brazil	160,204 (159,420-160,988)	7.5 (7.46-7.54)
Chile	18,531 (18,264-18,798)	9.6 (9.5-9.6)
Colombia	45,619 (45,200-46,038)	9.0 (8.9-9.1)
Ecuador	17,552 (17.292-17,812)	10.0 (9.9-10.1)
Falkland Islands	1 (0-3)	2.6 (0-7.7)
French Guiana	160 (135-1859)	5.5 (4.6-6.4)
Guyana	770 (716-824)	9.7 (9.0-10.4)
Paraguay	7785 (7612-7958)	11.8 (11.5-12.1)
Peru	38,466 (38,082-38,850)	11.5 (11.4-11.6)
Suriname	548 (502-594)	9.0 (8.2-9.8)
Uruguay	3437 (3322-3552)	10.0 (9.7-10.3)
Venezuela	31,343 (30,996-31,690)	11.0 (10.9-11.1)

selective abortions have had almost no impact on actual LB prevalence of DS. Only in a few countries/territories in this region have selective abortions had a significant impact, particularly in Cuba and French Overseas Territories (Figure 1).

The actual LB prevalence of DS in Latin America and the Caribbean was estimated at 18.4 per 10,000 LBs (95% CI: 18.2-18.7) for the period 2016 to 2020, which is significantly higher than in comparison regions. Recent estimates in these regions range from 6.9 per 10,000 LBs in New Zealand to 13.2 per 10,000 LBs in the United States (Figure 2). In Europe, only 2 countries—Malta, where abortion is illegal, and Ireland, where abortion regulations were very restrictive until 2019—have recent estimates of actual LB prevalence of DS higher than 15 per 10,000. In Latin America and the Caribbean, only 12 out of 50 countries (or territories) had an estimated actual LB prevalence lower than 15 per 10,000 LBs in recent years, and in 11 of these, this was primarily due to the impact of selective abortions. The estimates of actual LB prevalence of DS for 2016 to 2020 in Latin America and the Caribbean range from as low as 4.4 per 10,000 LBs in Cuba (95% CI: 3.2-5.7) to, if excluding very small countries/territories, 25.0 per 10,000 LBs in Haiti (95% CI: 23.1-26.9). This wide range reflects differences in maternal age across the general population, as well as varying approaches to prenatal screening and selective abortions.

Survival

Life expectancy can be measured as either mean or median. Mean life expectancy refers to the average number of years an individual is expected to live, whereas median life expectancy is the age at which half of a birth cohort is expected to have passed away. In the existing literature, the term "life expectancy" is sometimes used without specifying whether it refers to the mean or median.

In Latin America and the Caribbean, the mean life expectancy in the general population is lower than in the comparison regions, such as the United States, but the gap has narrowed over the past few decades (Supplemental Methods 2C). The same trend applies to the survival of people with DS (Supplemental Methods 2C). Although the estimated mean life expectancy for DS remains lower in Latin America and the Caribbean (48 years in 2020) than in the United States (54 years in 2020), there has been a rapid increase in Latin America and the Caribbean since 1985. However, there are significant differences within the region. On one hand, Cuba's estimated mean life expectancy for people with DS is now more favorable than that in the United States (Supplemental Methods 2C). On the other hand, in Haiti, the estimated mean life expectancy for people with DS remains much lower, although it has steadily increased from 3 years in 1950 to 27 years in 2020.

Population size and population prevalence of DS

The developments in the size of birth cohorts of children with DS, along with changes in the survival rates of people with DS, shape the development of both the population size and the population prevalence of DS. In contrast to the comparison areas, selective abortions in Latin America and the Caribbean have had a limited impact. The survival of people with DS is less favorable than in the comparison areas; however, survival in Latin America and the Caribbean has shown rapid improvement over the past few decades. The net effect of these developments is a rapid growth in the population size of DS, estimated at 574,092 (95% CI: 572,607-575,577) in 2020, which is 2.3 times larger than it was in 1990. By comparison, in the United States, the population of individuals with DS increased by approximately 1.4 times (between 1990 and 2018), in Australia and Canada by 1.3 times (between 1990 and 2020), and in New Zealand by 1.2 times (between 1990 and 2020).[1,4,5,19] In Europe (excluding the former Eastern bloc), the overall population of people with DS was even 1% lower in 2015 than in 1990.[3]

As a consequence of the aforementioned developments, the population prevalence of DS in Latin America and the Caribbean, estimated at 8.8 per 10,000 inhabitants (95% CI: 8.78-8.83), is significantly higher than in the comparison areas, where the estimates range from 5.2 per 10,000 inhabitants in Australia (as of 2020) to 6.7 per 10,000 inhabitants in the United States (as of 2018).[1,4,19] Within Latin America and the Caribbean, there are significant differences between countries/territories, with estimates of population prevalence of DS ranging from 2.6 per 10,000 inhabitants to 15.7 per 10,000 inhabitants. Details by region and country/territory can be found in Table 1 and in Supplemental Findings 1 and 2. In Supplemental Findings 3, we present a comparison with earlier estimates by the Global Burden of Disease Study.[20]

In 1950, in Latin America and the Caribbean, as well as in the comparison regions, DS was primarily a childhood condition, with the vast majority of individuals with DS being younger than 20 years of age, and very few reaching 40 years or older (Figure 3).[1,3-5,19] In the comparison regions, because of the improved survival rates of children and adults with DS, along with the reducing impact of selective terminations on the size of birth cohorts of children with DS, the percentage of individuals under 20 years of age decreased to between 43% (United States as of 2018) and 28% (Europe excluding the former East Bloc, as of 2015).[1,3,19] At the same time, the percentage of individuals aged 40 years and older increased to between 26% (United States as of 2018) and 41% (Europe excluding the former East Bloc as of 2015). In Latin America and the Caribbean, the survival of individuals with DS is improving, although it remains less favorable than in the comparison regions.

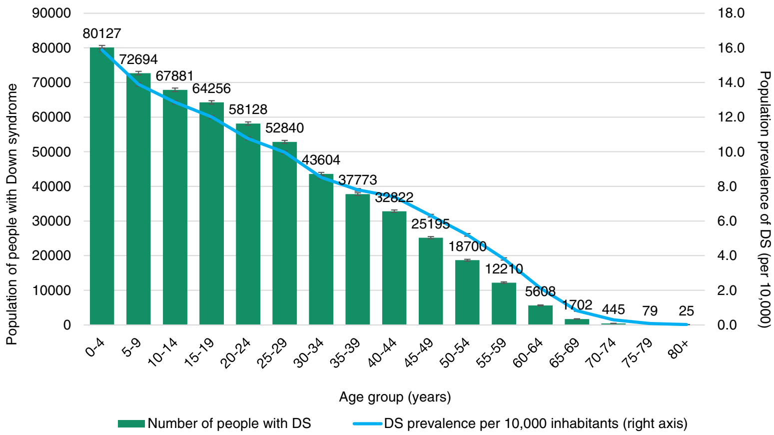

Figure 4 **Estimated people with DS in Latin America and the Caribbean by age group as of 2020.** DS, Down syndrome.

Additionally, the reduction in the size of birth cohorts of children with DS because of selective terminations is minimal. As a result, the percentage of individuals with DS under 20 years of age, estimated at 50%, is higher than in the comparison countries (Figures 3 and 4). Conversely, the percentage of individuals aged 40 and older, estimated at 17%, is lower than in the comparison countries (Figures 3 and 4).

Limitations of the method

All estimates in this study are based on modeling, and several assumptions had to be made. We assumed that in countries/territories with very restrictive abortion laws, the reduction due to selective abortions of children with DS would be 0 or near 0. However, it is possible that some selective abortions did occur, which could have led to some overestimation in our model of the actual LB prevalence of DS in these countries/territories. To validate our assumption of near 0 reduction, we compared our estimates of nonselective prevalence of DS with counts of liveborn children in several South American countries, and these aligned.[8-10] Subsequently, we modeled the likely reduction in the 18 countries/territories that allowed elective terminations for fetal impairment.[11] Reliable birth defect surveillance data were available for only 4 of these countries/territories.[10,21,22] Our assumptions for modeling reduction in the other 14 countries/territories—of which 11 were very small—are explained in Supplemental Methods 1B. Although we believe that our assumptions are reasonable, they may still be subject to some error, which could result in either some overestimation or underestimation of the LB prevalence of DS in these countries/territories.

Constructing DS-specific survival curves by year of birth and by country/territory, while accounting for the significant differences in the development of health care systems, was a challenge, and the modeled curves might differ from reality. However, we followed an extensive validation approach, presented in Supplemental Methods 3. The model predictions for the number of people with DS by age group align with actual counts in 4 comparison countries (United Kingdom, France, Australia, and the United States) and, to a large extent, also with counts in Cuba and Argentina. Additionally, the model's predictions for the age distribution at death for people with DS in different countries/territories in Latin America and the Caribbean align with this distribution in national statistics and capture the large differences between countries/territories, as well as changes over time.[18]

We have not examined potential differences between subgroups within countries or territories (for instance, ethnicity or income). We assumed that such differences would level out in the estimation of people with DS for the country or territory as a whole. However, intracountry differences in survival may not fully cancel out because the model includes some nonlinear steps, and certain steps are conditional upon specific thresholds.

The total for the larger geographical regions has now been constructed by aggregating the results from each country or territory, which we consider to be the more precise method because it accounts for differences in child mortality between countries and territories. However, an alternative approach would be to estimate the results for the larger regions directly, based on the average child mortality in those regions. If the results of these 2 methods are similar, it would suggest that the existing large differences in survival between countries do, indeed, level out, which, in turn, supports the assumption that intracountry differences are also likely to level out. Using this alternative estimation method resulted in an estimated number of people with DS in Latin America and the Caribbean that is only 2% lower than the original aggregated estimate. For the Caribbean specifically, the estimate is 0.4% higher; for Central America and Mexico, 0.2% lower; and for South America, 3% lower. These small differences support our assumption that disparities, both between and within countries, tend to level out

Finally, we assumed that there was limited migration of people with DS into or from the different countries/

territories, and that their net migration was 0. Earlier studies demonstrated that, in the United States, a country with significant immigration, the effect of migration of people with DS on the population size of DS was small, both for the country as a whole and for individual states.[1,2] However, it cannot be ruled out that the large migration flows in the Latin American and Caribbean region may have included individuals with DS, particularly if they migrated as young children with their parents. If significant numbers were involved—such as may have been the case with refugees from Cuba in the past—this could have had an impact on the population size of DS. We did not model this because it remains uncertain and difficult to quantify.

Is the population size of DS contracting?

Because of selective terminations, the population of people with DS has started to decline in Australia, New Zealand, and many European countries.[3,4] In Latin America and the Caribbean, according to our model, the overall population of people with DS is still growing significantly, as a result of the sharply increasing survival rate estimates for children with DS in this region in recent decades, combined with relatively few selective abortions. Only in 5 countries/territories within this region has the overall population of people with DS begun to decrease in recent years (Supplemental Findings 1). In 2 of these countries (Cuba and Guadeloupe), the population would have increased if selective abortions had never occurred. In the other 3 countries/territories, selective abortions play no role (Montserrat) or only a partial role (Martinique and Puerto Rico) in the downward population trend for DS. The decrease in the overall population of people with DS in these 3 countries/territories is entirely or largely explained by a significant decline in the total number of children born in the general population. If, in the future, prenatal screening for DS becomes more widely available in Latin America and the Caribbean, and if elective terminations for fetal impairment are legalized in more countries, a stronger slowdown in the growth of the overall population of people with DS would be expected, and potentially, in the long term, the population would begin to contract.

Data Availability

The authors will supply data and materials individually upon request.

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Author Contributions

Conceptualization: G.d.G., F.B., B.G.S.; Data Curation: G.d.G.; Formal Analysis: G.d.G.; Investigation: G.d.G.; Methodology: G.d.G.; Project Administration: G.d.G., B.G.S.; Resources: G.d.G.; Software: G.d.G.; Supervision: G.d.G., B.G.S.; Validation: G.d.G.; Visualization: G.d.G., F.B.; Writing-original draft: G.d.G., B.G.S.; Writing-review and editing: G.d.G., F.B., B.G.S.

ORCIDs

Gert de Graaf: https://orcid.org/0000-0002-4990-5711
Brian G. Skotko: https://orcid.org/0000-0002-5232-9882

Ethics Declaration

Only publicly available data were used for this study. All of the data were deidentified and did not qualify as human subjects research. As such, IRB approval was not necessary.

Conflict of Interest

Brian G. Skotko occasionally consults on the topic of Down syndrome through Gerson Lehrman Group. He receives remuneration from Down syndrome non-profit organizations for speaking engagements and associated travel expenses. Within the past 2 years, he has received research funding from AC Immune and LuMind Research Down Syndrome Foundation to conduct clinical trials for people with Down syndrome. Brian G. Skotko is occasionally asked to serve as an expert witness for legal cases where

Down syndrome is discussed. Brian G. Skotko serves in a non-paid capacity on the Honorary Board of Directors for the Massachusetts Down Syndrome Congress and the Professional Advisory Committee for the National Center for Prenatal and Postnatal Down Syndrome Resources. Brian G. Skotko has a sister with Down syndrome.

Gert de Graaf had a daughter with DS, who passed away in 2005 at the age of 15. He works as science and education officer at the Dutch Down Syndrome Foundation, a nonprofit organization.

Frank Buckley serves as CEO of Down Syndrome Education International and Down Syndrome Education USA, nonprofits engaged in research and support for young people with Down syndrome. He had a sister with Down syndrome.

Additional Information

The online version of this article (https://doi.org/10.1016/j.gim.2025.101621) contains supplemental material, which is available to authorized users.

References

1. de Graaf G, Buckley F, Skotko BG. Estimation of the number of people with Down syndrome in the United States. *Genet Med*. 2017;19(4):439-447. http://doi.org/10.1038/gim.2016.127
2. de Graaf G, Buckley F, Dever J, Skotko BG. Estimation of live birth and population prevalence of Down syndrome in nine U.S. states. *Am J Med Genet A*. 2017;173(10):2710-2719. http://doi.org/10.1002/ajmg.a.38402
3. de Graaf G, Buckley F, Skotko BG. Estimation of the number of people with Down syndrome in Europe. *Eur J Hum Genet*. 2021;29(3):402-410. http://doi.org/10.1038/s41431-020-00748-y
4. de Graaf G, Skladzien E, Buckley F, Skotko BG. Estimation of the number of people with Down syndrome in Australia and New Zealand. *Genet Med*. 2022;24(12):2568-2577. http://doi.org/10.1016/j.gim.2022.08.029
5. de Graaf G, LaChance L, Buckley F, Skotko BG. Estimation of the number of people with Down syndrome living in Canada. *Genet Med*. 2025;27(7):101422. http://doi.org/10.1016/j.gim.2025.101422
6. de Graaf G, Buckley F, Skotko BG. Estimates of the live births, natural losses, and elective terminations with Down syndrome in the United States. *Am J Med Genet A*. 2015;167A(4):756-767. http://doi.org/10.1002/ajmg.a.37001
7. Morris JK, Mutton DE, Alberman E. Revised estimates of the maternal age specific live birth prevalence of Down's syndrome. *J Med Screen*. 2002;9(1):2-6. http://doi.org/10.1136/jms.9.1.2
8. Martini J, Bidondo MP, Duarte S, Liascovich R, Barbero P, Groisman B. Birth prevalence of Down syndrome in Argentina. *Salud Colect*. 2019;15:e1863. http://doi.org/10.18294/sc.2019.1863
9. Campaña H, Pawluk MS, López Camelo JS, del Eclamc G de E. Prevalencia al nacimiento de 27 anomalías congénitas seleccionadas. 7 regiones geográficas de la Argentina. Accessed December 17, 2024. https://ri.conicet.gov.ar/handle/11336/97955
10. Annual report. International Clearinghouse for Birth Defects. Accessed December 17, 2024. http://www.icbdsr.org/resources/annual-report/
11. Global Abortion Policies Database: is designed to strengthen global efforts to eliminate unsafe abortion. World Health Organization. Accessed December 17, 2024. https://abortion-policies.srhr.org/
12. Registro nacional de personas con discapacidad. Personas con Certificado Único de Discapacidad con condiciones de salud vinculadas al Síndrome de Down y/o anomalías cromosómicas no especificadas. Agencia Nacional de Discapacidad. Accessed December 27, 2024. https://www.argentina.gob.ar/andis/estadistica-e-investigacion-social/informes
13. Lantigua A, Portuondo M, Collazo T, Lardoeyt R. Epidemiology of prenatal genetic and environmental factors of mental retardation in Cuba. *MEDICC Rev*. 2008;10(1):29-36. http://doi.org/10.37757/MR2008.V10.N1.9
14. Alexander M, Ding Y, Foskett N, Petri H, Wandel C, Khwaja O. Population prevalence of Down's syndrome in the United Kingdom. *J Intellect Disabil Res*. 2016;60(9):874-878. http://doi.org/10.1111/jir.12277
15. Clift AK, Coupland CAC, Keogh RH, Hemingway H, Hippisley-Cox J. COVID-19 mortality risk in Down syndrome: results from a cohort study of 8 million adults. *Ann Intern Med*. 2021;174(4):572-576. http://doi.org/10.7326/M20-4986
16. Tuppin P, Barthelemy P, Debeugny G, Rachas A. A French nationwide study compared various conditions and healthcare use of individuals <65 years with a Down's syndrome to those without. *Sci Rep*. 2023;13(1):21865. http://doi.org/10.1038/s41598-023-49102-4
17. Rubenstein E, Tewolde S, Michals A, et al. Alzheimer dementia among individuals with Down syndrome. *JAMA Netw Open*. 2024;7(9):e2435018. http://doi.org/10.1001/jamanetworkopen.2024.35018
18. Raw data files of the WHO Mortality Database. World Health Organization. Accessed September 17, 2019. https://www.who.int/healthinfo/statistics/mortality_rawdata/en/
19. de Graaf G, Buckley F, Skotko B. People living with Down syndrome in the USA: BIRTHS AND POPULATION. | https:. https://go.downsyndromepopulation.org/usa-factsheet
20. Global Burden of Disease Collaborative Network. Global burden of disease Study 2021 (GBD 2021) results. Institute for Health Metrics and Evaluation. Accessed February 2, 2025. https://vizhub.healthdata.org/gbd-results
21. Méndez-Rosado LA, Hechavarría-Estenoz D, de La Torre ME, et al. Current status of prenatal diagnosis in Cuba: causes of low prevalence of Down syndrome. *Prenat Diagn*. 2014;34(11):1049-1054. http://doi.org/10.1002/pd.4421
22. Prevalence charts and tables. European Commission. Accessed December 12, 2024. https://eu-rd-platform.jrc.ec.europa.eu/eurocat/eurocat-data/prevalence_en